

ABSTRACT.

Iceland

Sparsely populated island in the North Atlantic, known for its stunning natural landscapes, including volcanoes, glaciers, and geothermal hot springs.

Most residents are concentrated in the capital, Reykjavik, and surrounding areas. It's essential to provide comfortable and sheltering living, due to the island's harsh climate and shortage of sunlight.

the task

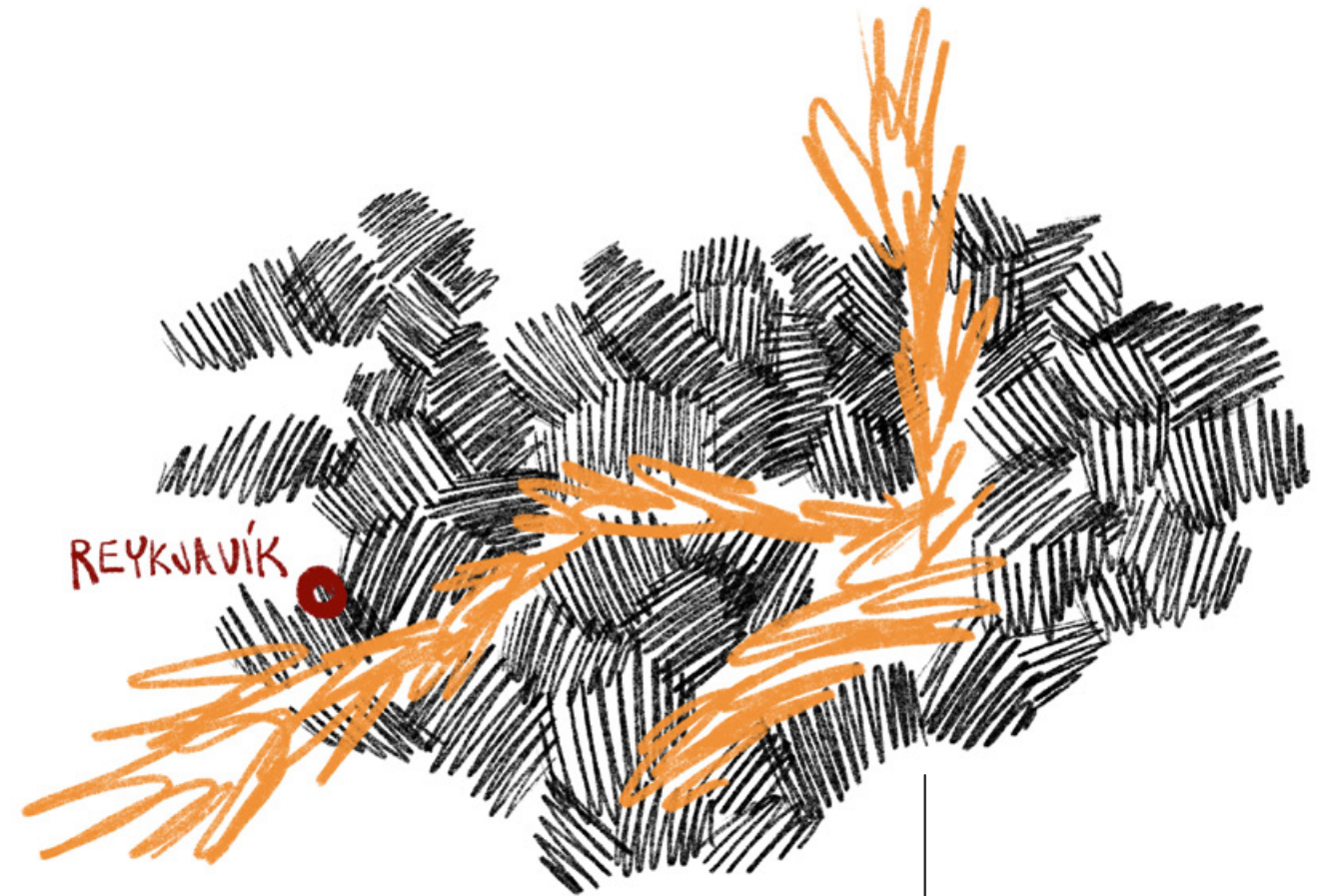
The goal is to translate the regional strategy and analysis of Iceland, Reykjavik, and Keldur area into a tangible, site-specific design that fosters a new, livable urban fabric. Explore innovative typologies that promote sustainability and resilience while ensuring the design is adapted to the unique landscape and topography of Keldur.

Highlight how your project responds to the distinctive landscape features, the local community, ecological systems, and the multiple aspects of sustainability in Reykjavik.

regional strategy

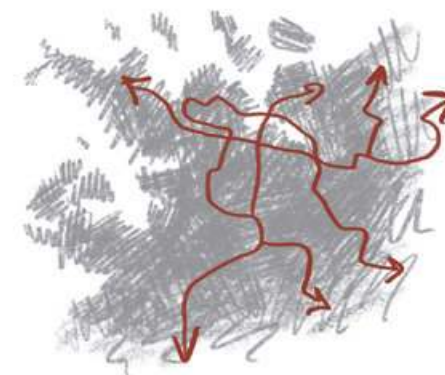
Mine regional strategy is based on the previous groupwork, consisting of the general Iceland, Reykjavík, Keldur area and analysis, plus strategy discussions.

An essential step is to enhance the infrastructure, because the city has very dysfunctional public transport. The municipality has planned to apply the BRT (bus rapid transit).

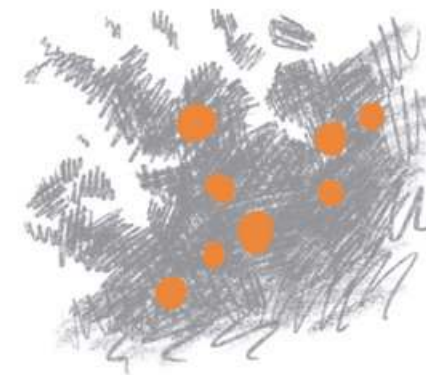


Iceland island diagram

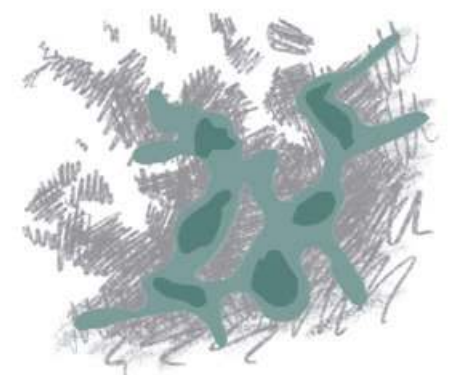
Reykjavík city diagram



ENHANCING
THE INFRASTRUCTURE



MAPPING 3RD
LANDSCAPES FOR
DENSIFYING



BUILDING WHILE
PRESERVING

SITE ANALYSIS

THE KELDUR AREA

gap in an urban tissue of very sprawled city

mostly south-facing slope

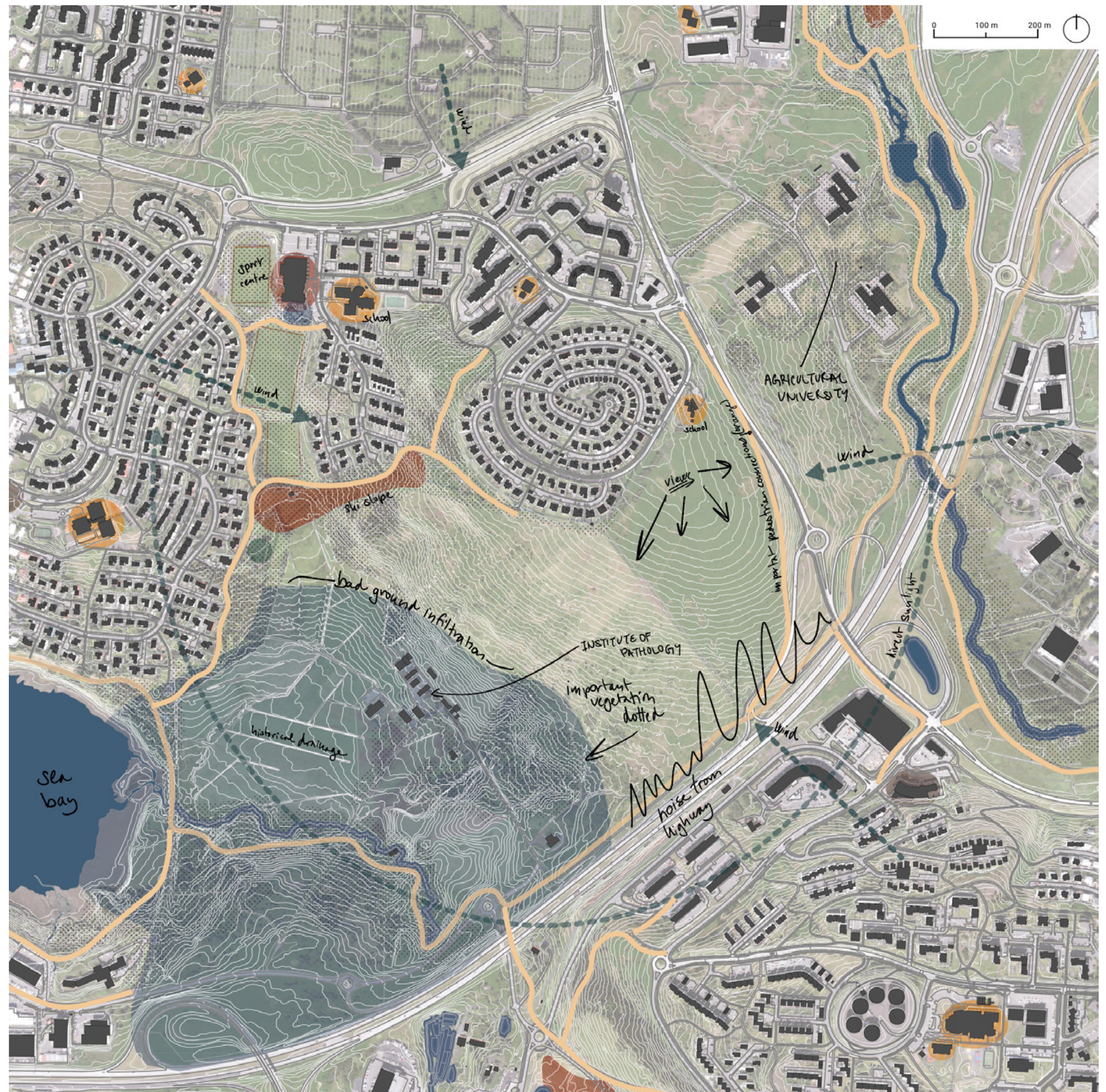
adjacent to sea bay on one end

cultural legacy - historical place for farming

slow development of vegetation

no big fauna, mostly birds and insects

the surrounding urban tissue is seemly designed without relation to the landscape

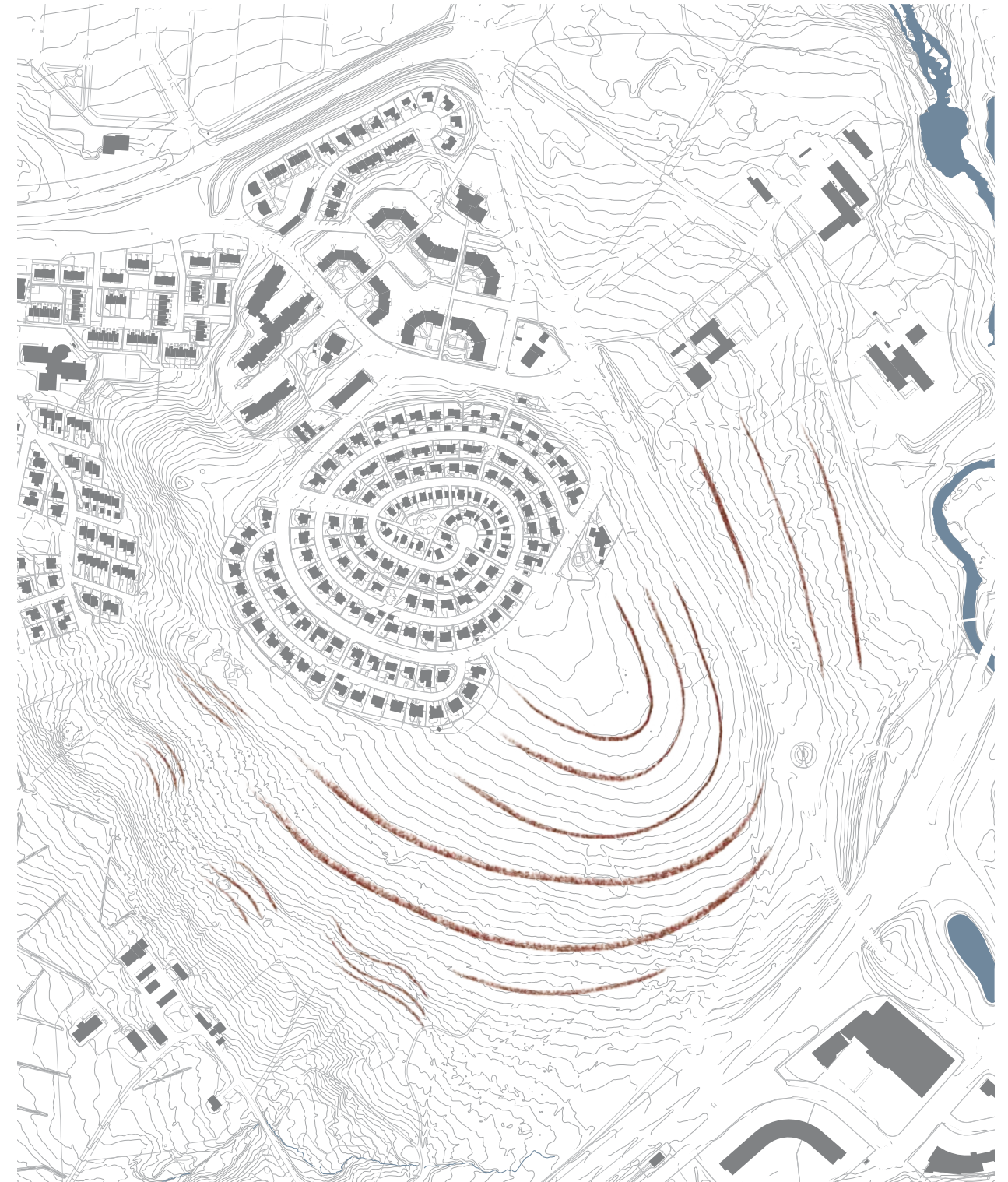


THE STORY OF ABSTRACTION

By the process of abstracting you're gaining just the essential things you need, but still in a relation and connection to the existing.

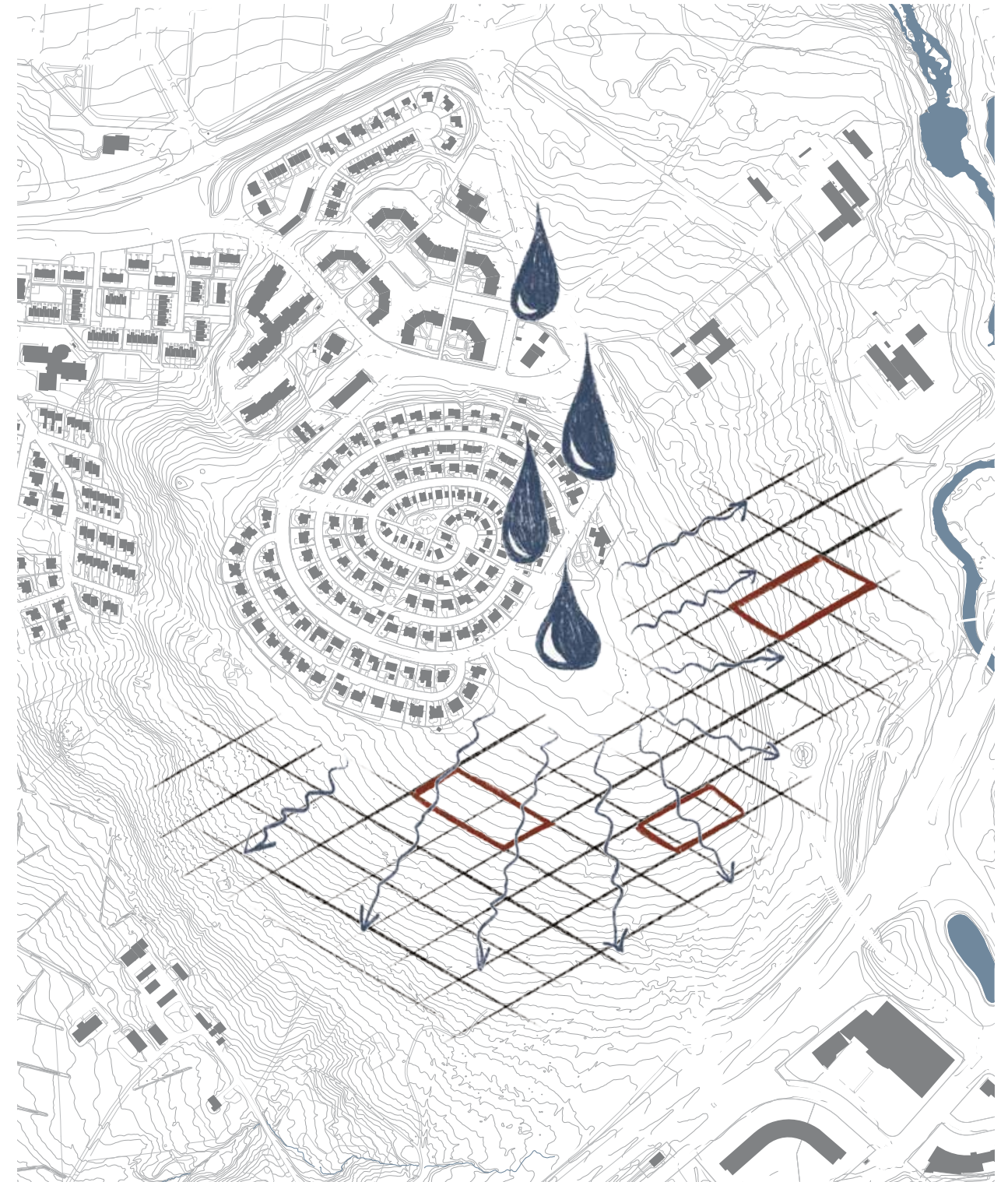
Basing the design on the landscape makes the place more readable for the visitors, and people can connected emotionally to it.

STRATEGY OF ABSTRACTING



the terrain

The design is interpreting the main shape features of the contour lines.



the vegetation

Existing vegetation, especially the valuable shrubs and trees, marked by darker green.

New areas that will create an ecological network based on that, marked with lighter green.

the areas for new development

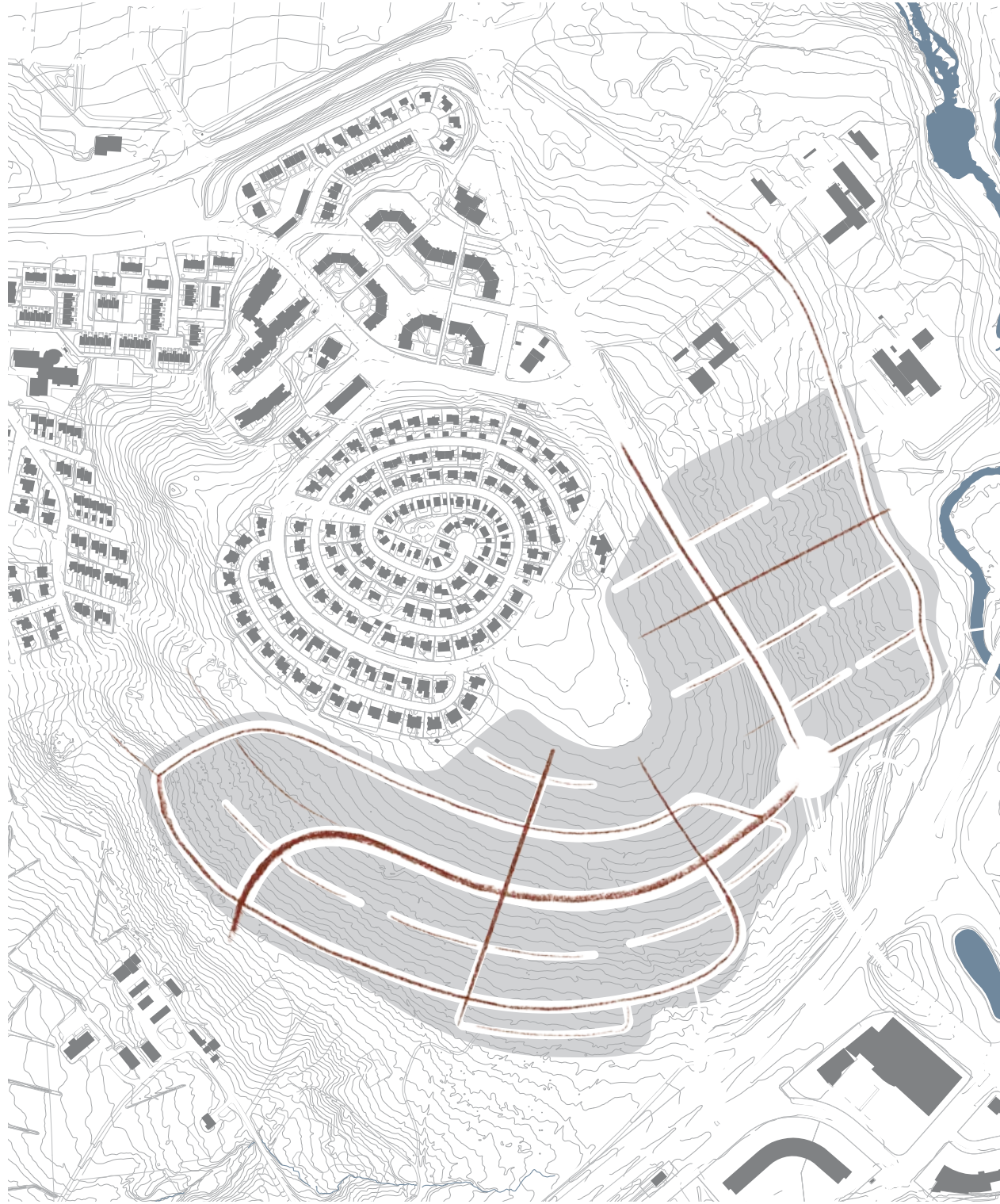
Marked with orange, together with the green network creating dense but breathable urban structure.

the aim of the ideal sun exposure

Determining the best angles of streets and therefore buildings, using the help of approx. 120° angle grid.

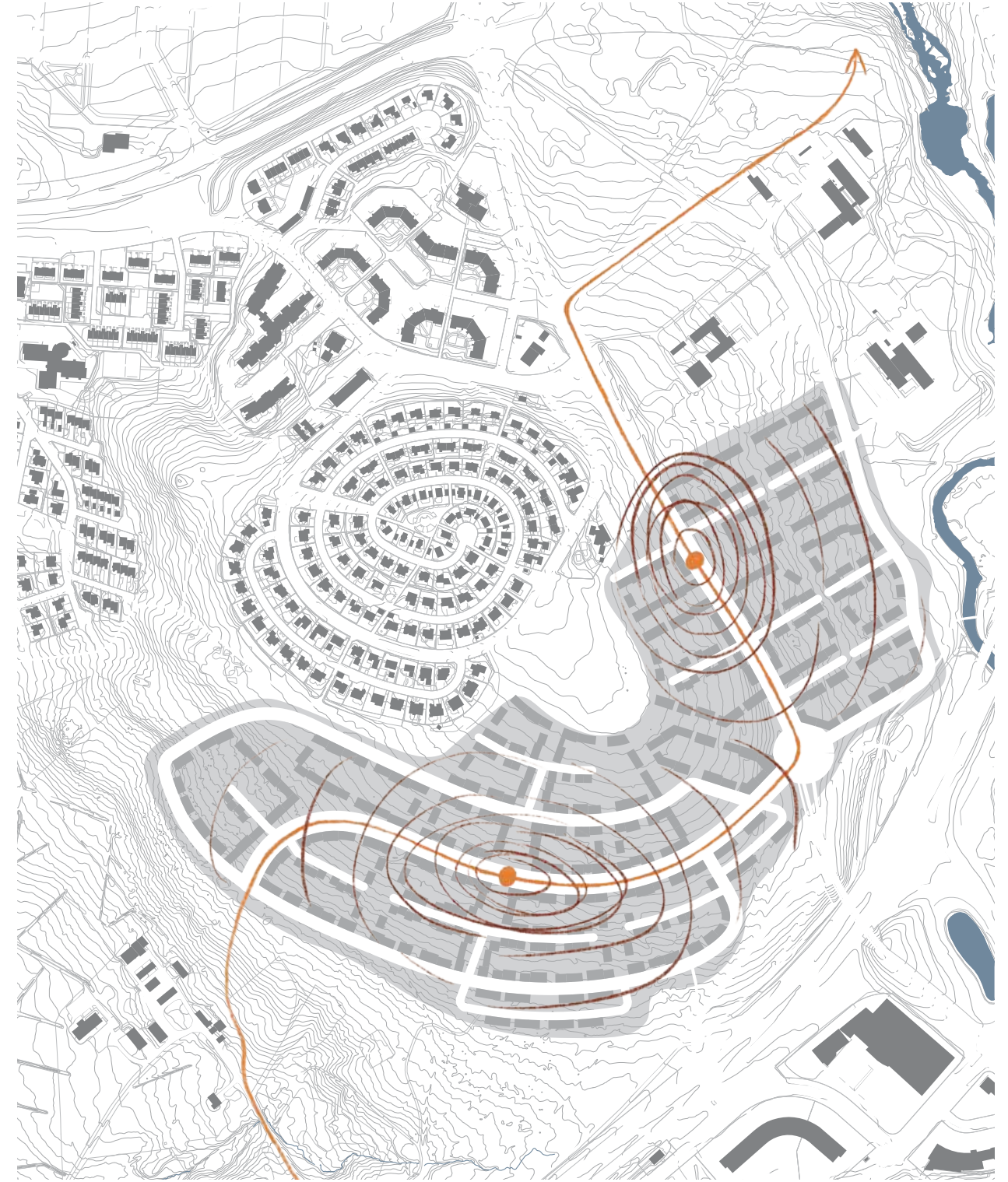
the centralization by the hill

Highlighting the feel of the terrain in the urban structure by radial streets. Also ideal for managing water and creating framed views from the top.



creating the street structure

The structure rises from all the previous aspect gently molded together. It makes the landscape readable for the visitors trough the process of abstraction.



BRT and the density

By placing the BRT line and it's stops, the density and the typologies filling the blocks are determined.

ABSTRACT.

The designed city part is centralized by the hilltop park, because I wanted to keep the top free and **offer connecting public space** to the already present adjacent neighborhood.

I was strongly inspired by streets in the old city center of Reykjavik, where you have a framed view uphill, crowned with the cathedral. **I'm using this principle by placing two axes** (created by pedestrian streets), that connect the BRT stops with the hilltop community center (building of sculptural architecture).

The aim was to create a compact fabric, as opposed to the sprawled structures around.

services

- 1/ COMMUNITY CENTER
- 2/ CARE FOR SENIOR CITIZENS
- 3/ MARKET HALL + OFFICES
- 4/ SCHOOL
- 5/ LIBRARY
- 6/ OFFICES
- 7/ MOBILITY HUB
- 8/ SWIMMING POOL
- 9/ CAFÉ / BISTRO / TOILETS
- 10/ WATCHTOWER AND BIRD OBSERVATORY

masterplan

view axis

BRT line

active pub.sp.

private gardens

added vegetation (green-belt)

existing valuable vegetation

paved public space



blue and green

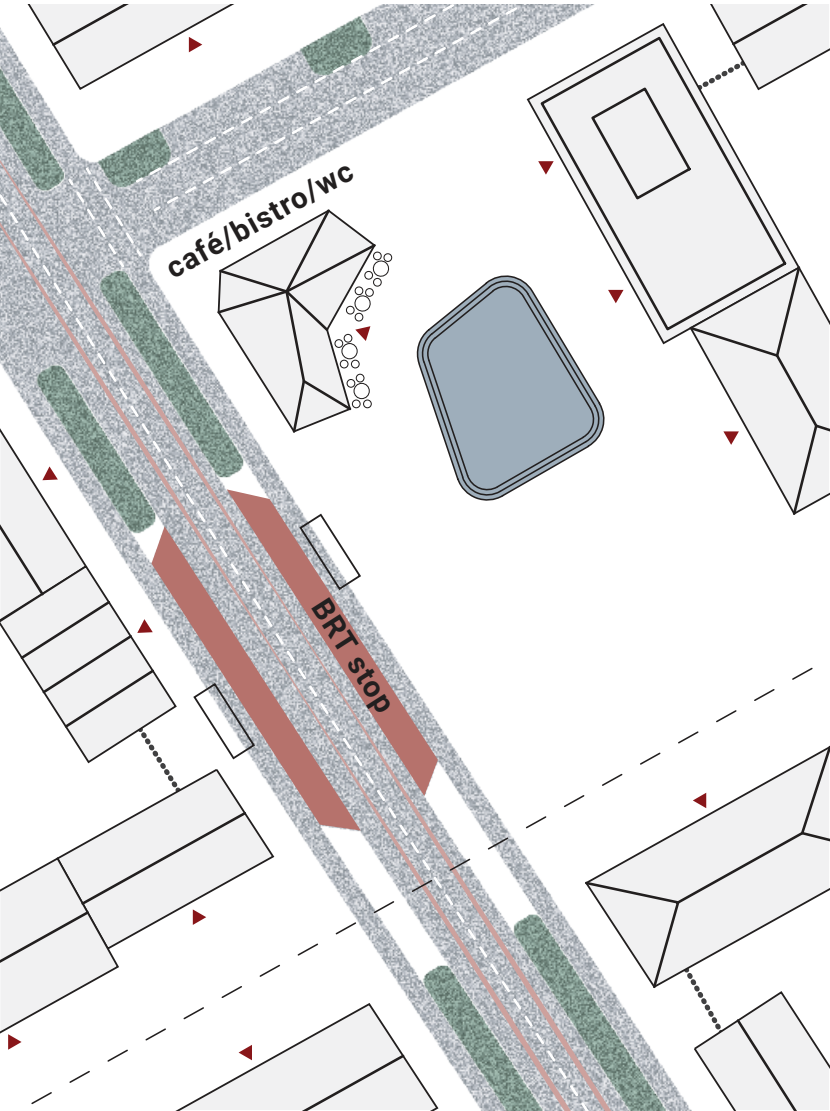
See how is the outer great green network connected to the inner-city econetwork, and the proposal of managing the rainwater locally.



- inner-city econetwork
- existing valuable vegetation
- added vegetation creating the green-belt econetwork
- green-belt econetwork
- rainwater distribution
- rainwater ponds



infrastructure



DETAIL PLAN CUTOUT

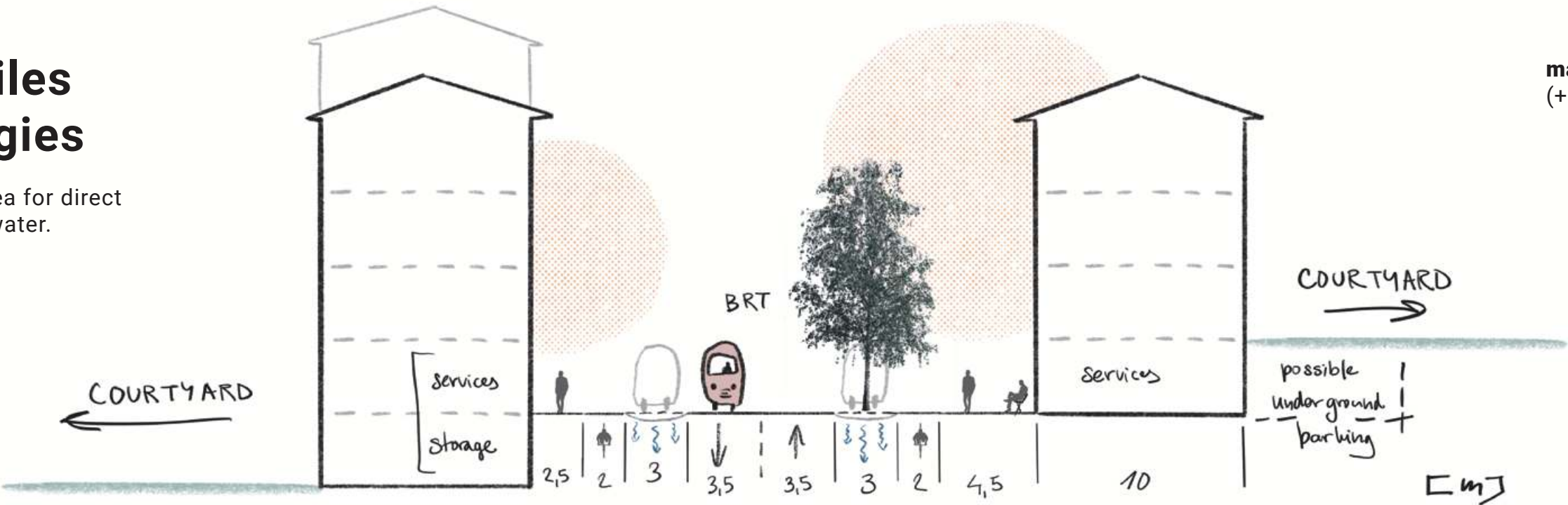
the sections of the street profiles are on the next page

-  BRT line
-  car-loop main roads
-  resident-only streets
-  connections to adjacent neighborhoods
-  mobility hubs

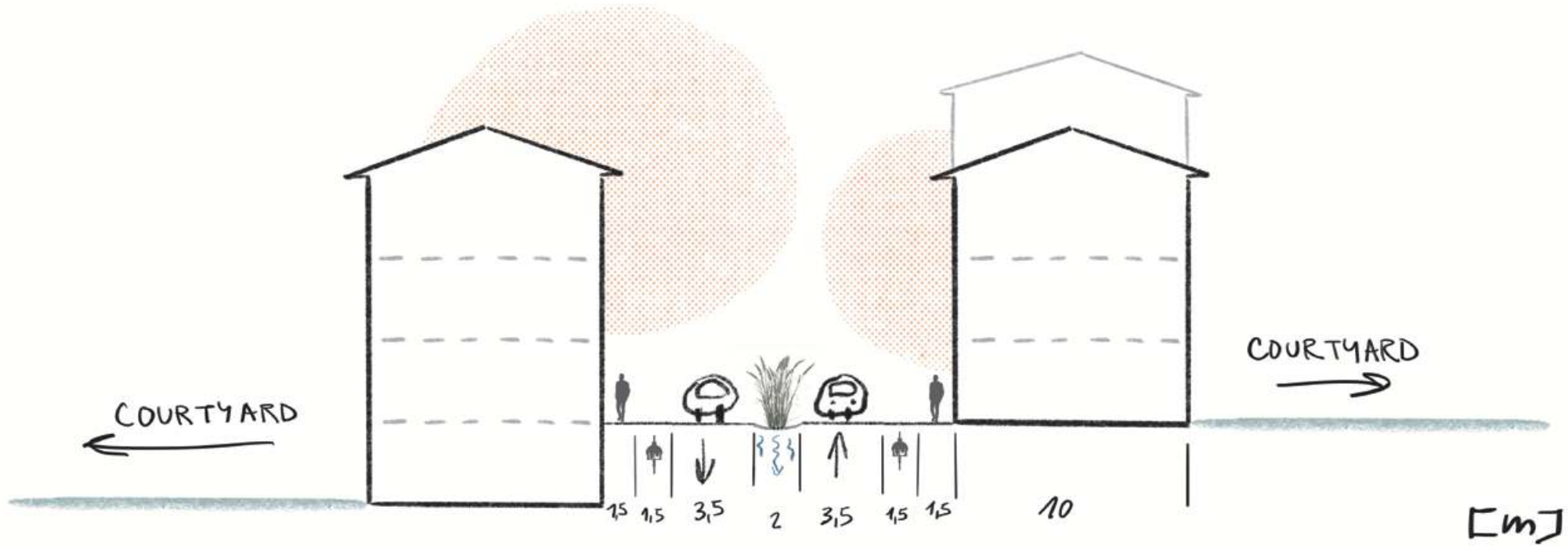


street profiles and typologies

Every street has an area for direct infiltration of the rainwater.



main street with BRT
(+ apartment buildings)



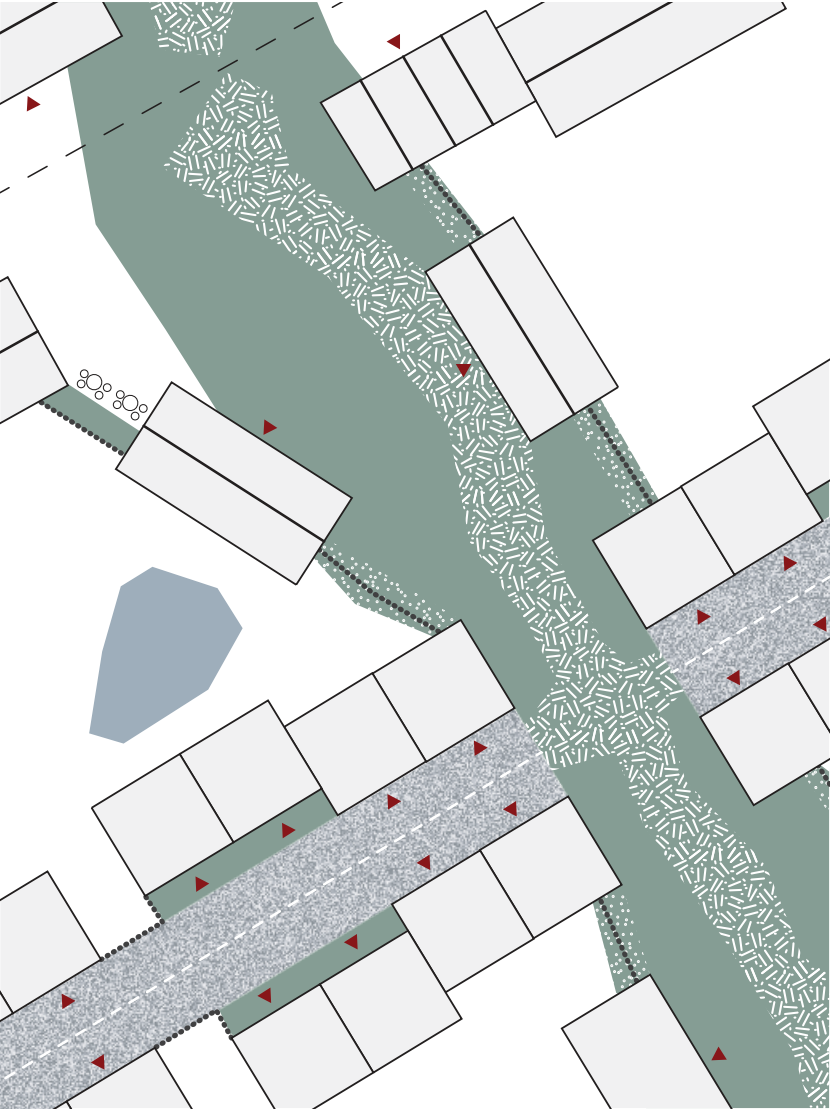
regular car street
(+ apartment buildings)



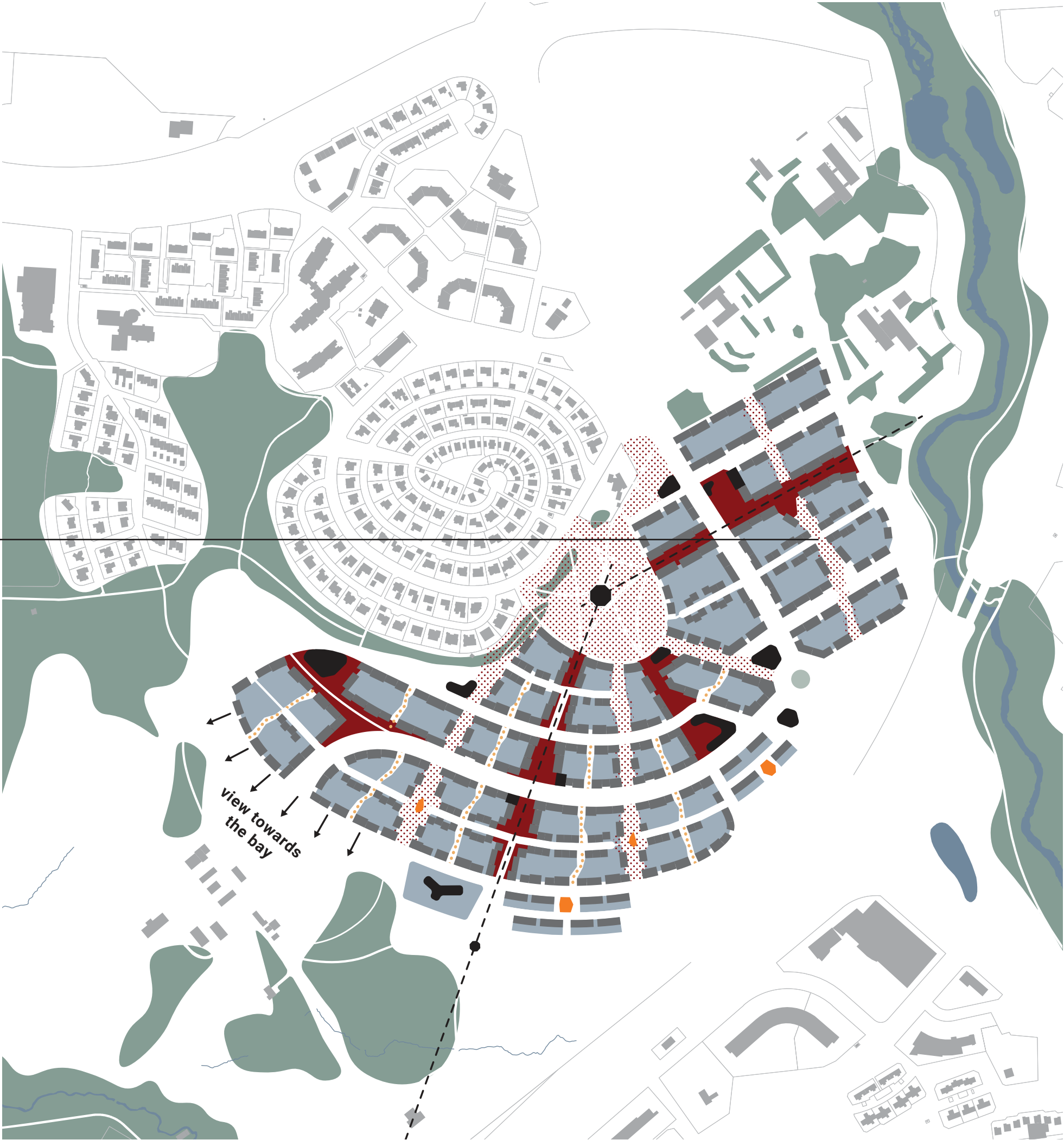
residential street
(+ row-houses)

created exterior spaces

Divided into 6 categories by accessibility. See the visualizations on the next pages.



- public_ACTIVE SIDEWALK
- public_MAIN CENTERS
- public_GREEN
- semipublic_MEETING POINTS
- semipublic_ALLEYS
- private_ENCLOSED



public_MAIN CENTERS (pedestrian streets, squares)

Two main axes connecting visually each quarter to the hilltop park and community center.

Functioning as a cozy pedestrian street, full of services and places to meet.



MASTERPLAN CUTOUT 1:2000



public_ACTIVE SIDEWALK (around public buildings, BRT street, local centers)

Buildings with services located on the street level attracting people and making the space alive.

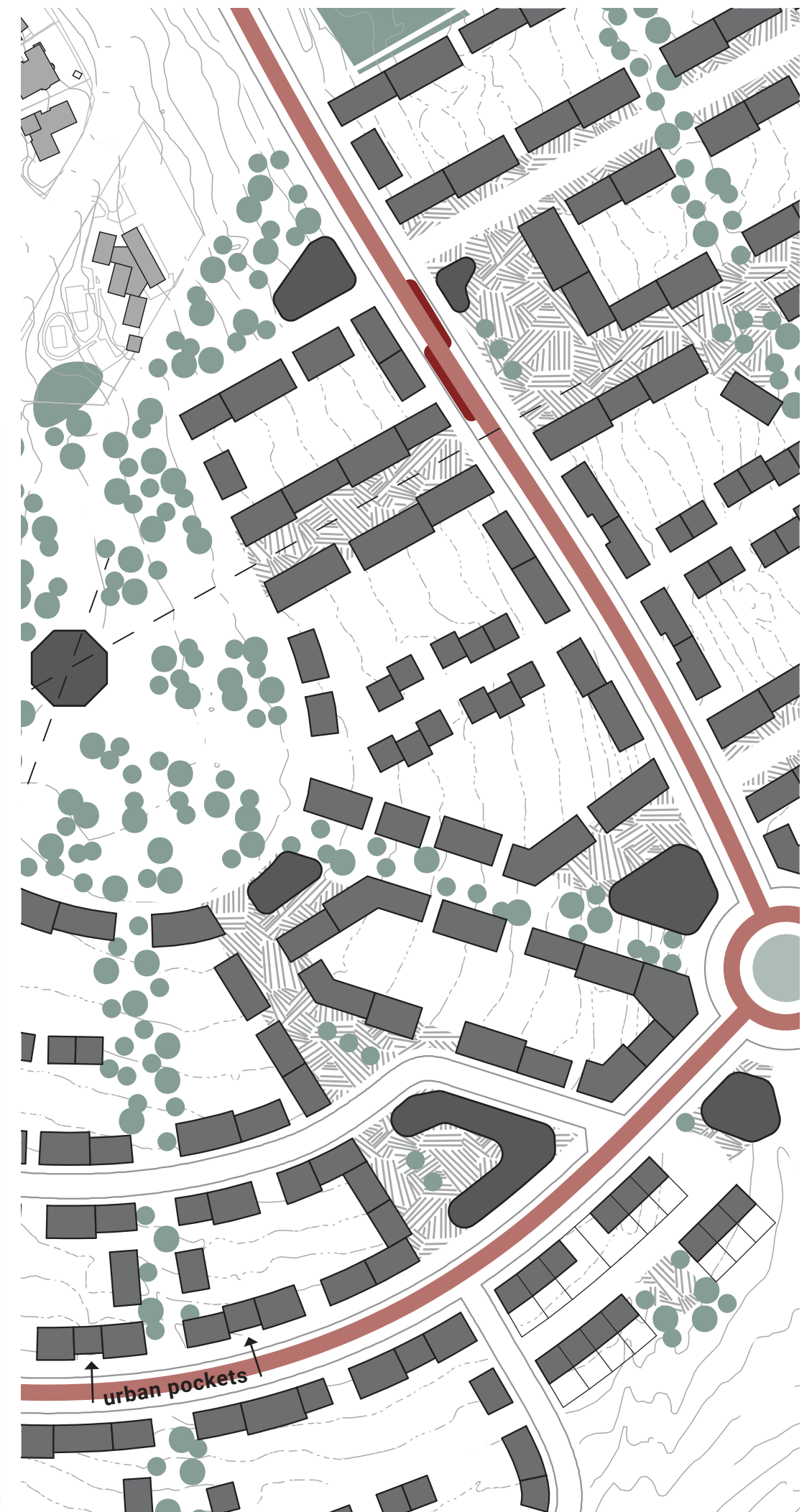
marked with orange glow

special feature: **URBAN POCKET**

Uneven line of facades on the sunny side of the street creating smaller warmer spaces, sheltered from the wind.



MASTERPLAN CUTOUT 1:2000



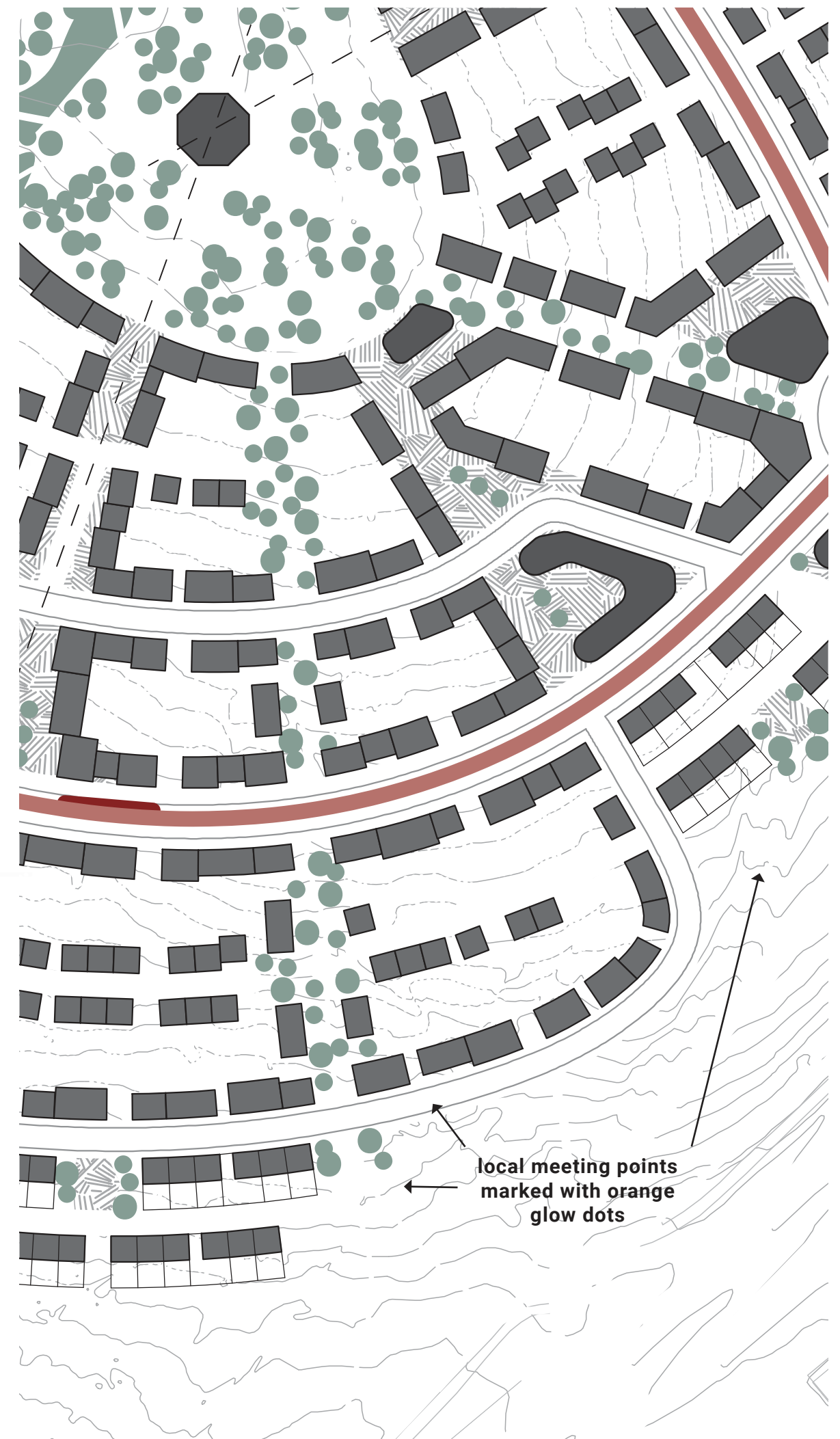
semipublic_LOCAL MEETING POINTS (connecting blocks, row-house neighborhoods)

A hangout spot with table and barbecue grill / fireplace, ideally sheltered.

Functioning as a shared place, opportunity for people not just from row-houses to crate a community.



MASTERPLAN CUTOUT 1:2000



public_GREEN

(corridors and parks)

Offering activities such as playgrounds or sport equipment, for recreation all year round.

Functioning as a intense green econetwork through the city.



semipublic_ALLEYS

(through the courtyards)

Green alleys divided from the courtyards by hedges and low fences / stonewalls, with occasional bench.

Functioning as an element of pedestrian permeability and a place for a calm stroll through the neighborhood.

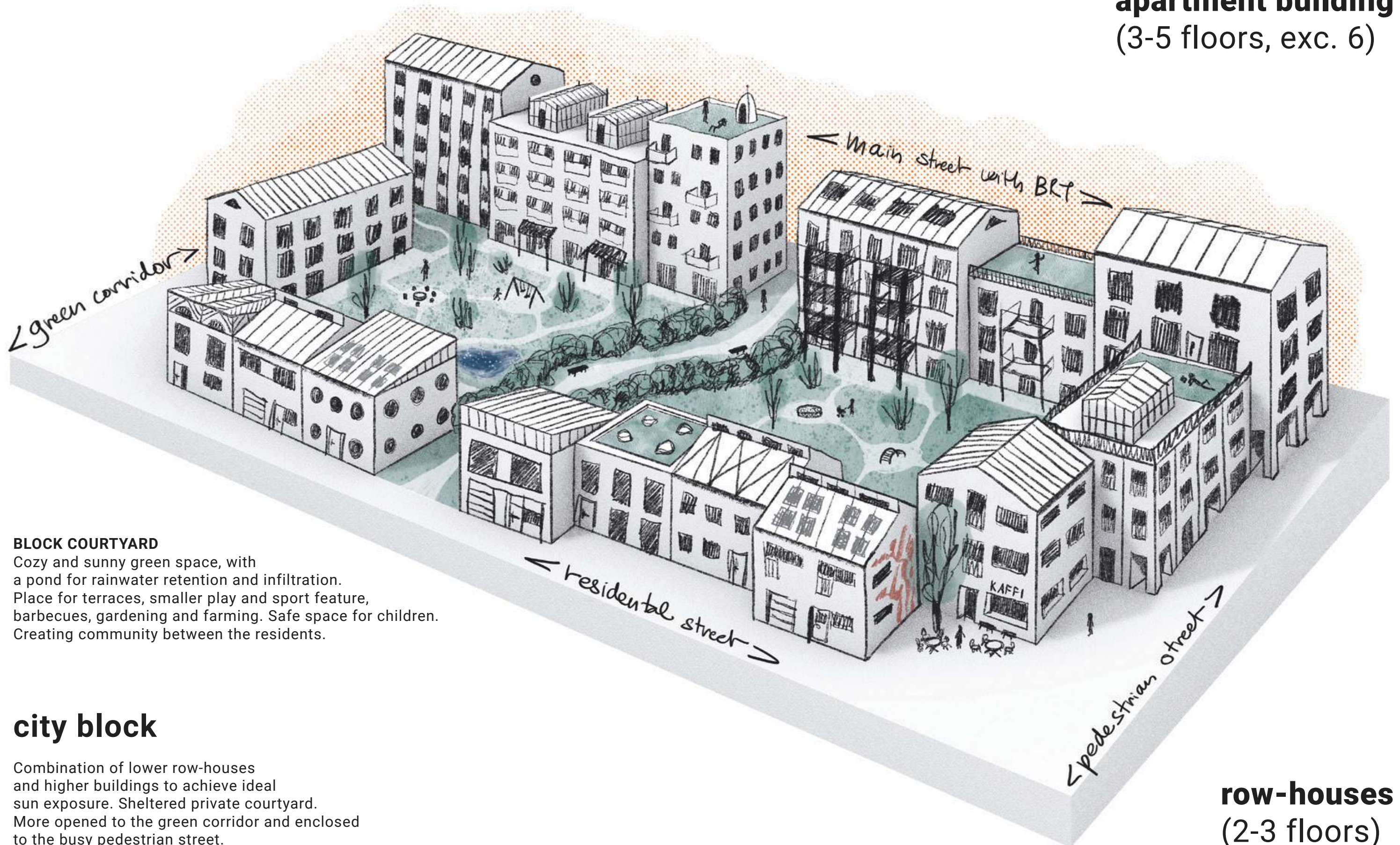
visible on the next page



private_ENCLOSED (block courtyards, row-house gardens)

TPOLOGIES:

apartment building
(3-5 floors, exc. 6)



BLOCK COURTYARD

Cozy and sunny green space, with a pond for rainwater retention and infiltration. Place for terraces, smaller play and sport feature, barbecues, gardening and farming. Safe space for children. Creating community between the residents.

city block

Combination of lower row-houses and higher buildings to achieve ideal sun exposure. Sheltered private courtyard. More opened to the green corridor and enclosed to the busy pedestrian street.

row-houses
(2-3 floors)